

EE 133 Homework 3

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Problem 1

For $E - E_C = 0.01$ eV and $m_n^* = 1.18m_0$, calculate the density of states $g_c(E)$ in units of $(\frac{1}{\text{eV} \cdot \text{cm}^3})$. (Hint: Value is between 10^{20} and 10^{21}).

Solution

- **Given:** $m_n^* = 1.18m_0$, $E - E_C = 0.01$ eV
- **Formula:**

$$g_c(E) = \frac{1}{2\pi^2} \left(\frac{2m_n^*}{\hbar^2} \right)^{3/2} \sqrt{E - E_C}$$

Calculation

Constants & conversions

$$\begin{aligned} m_n^* &= 1.18 (9.11 \times 10^{-31} \text{ kg}) = 1.075 \times 10^{-30} \text{ kg}, \\ E - E_C &= 0.01 \text{ eV} = 1.602 \times 10^{-21} \text{ J}. \end{aligned}$$

Density of states (SI units)

$$\begin{aligned} g_c(E) &= \frac{1}{2\pi^2} \left(\frac{2(1.075 \times 10^{-30})}{(1.054 \times 10^{-34})^2} \right)^{3/2} \sqrt{1.602 \times 10^{-21}} \\ &\approx 5.45 \times 10^{45} \text{ J}^{-1} \cdot \text{m}^{-3}. \end{aligned}$$

Unit conversion ($\text{eV}^{-1} \cdot \text{cm}^{-3}$)

$$\begin{aligned} g_c(E) &= (5.45 \times 10^{45}) (1.602 \times 10^{-19}) (10^{-6}) \\ &\approx 8.74 \times 10^{20} \text{ eV}^{-1} \cdot \text{cm}^{-3}. \end{aligned}$$

Final answer

$g_c(E) \approx 8.74 \times 10^{20} \text{ eV}^{-1} \cdot \text{cm}^{-3}$

Problem 2

When $E > E_F + 3k_B T$, we can approximate the Fermi factor $f(E) = \frac{1}{1 + e^{(E-E_F)/k_B T}}$ with the Boltzmann factor $B(E) = e^{-(E-E_F)/k_B T}$.

- (a) Derive an expression for the percent error $[B(E) - f(E)]/f(E)$ for arbitrary temperature $T > 0$.
- (b) Determine the actual percent error for $E = E_F + 3k_B T$.
- (c) Determine the actual percent error for $E = E_F + 4k_B T$.
- (d) Determine the actual percent error for $E = E_F + 5k_B T$.

Solution

Given:

$$f(E) = \frac{1}{1 + e^{(E-E_F)/k_B T}}, \quad B(E) = e^{-(E-E_F)/k_B T}.$$

- (a) **Percent error (relative to $f(E)$):**

$$\text{Error} = \frac{B(E) - f(E)}{f(E)}.$$

Let $x = \frac{E-E_F}{k_B T}$ so that $f(E) = \frac{1}{1+e^x}$ and $B(E) = e^{-x}$. Then

$$\begin{aligned} \text{Error} &= \frac{e^{-x} - \frac{1}{1+e^x}}{\frac{1}{1+e^x}} = \left(e^{-x} - \frac{1}{1+e^x} \right) (1+e^x) \\ &= e^{-x}(1+e^x) - 1 = (e^{-x} + 1) - 1 = e^{-x}. \end{aligned}$$

Therefore,

$$\% \text{ Error} = e^{-\frac{E-E_F}{k_B T}} \times 100\%.$$

- (b) For $E = E_F + 3k_B T$ ($x = 3$):

$$\% \text{ Error} = e^{-3} \times 100\% \approx \mathbf{4.98\%}.$$

- (c) For $E = E_F + 4k_B T$ ($x = 4$):

$$\% \text{ Error} = e^{-4} \times 100\% \approx \mathbf{1.83\%}.$$

- (d) For $E = E_F + 5k_B T$ ($x = 5$):

$$\% \text{ Error} = e^{-5} \times 100\% \approx \mathbf{0.67\%}.$$

Problem 3

The condition $(E_v + 3kT) \leq E_F \leq (E_c - 3kT)$ separates two different regimes.

- (a) What are they?
- (b) Write the 2 expressions for electron density and 2 expressions for hole density valid in this regime.

Solution

- (a) **Two regimes**

This condition separates **degenerate** and **non-degenerate** semiconductors:

- **Non-degenerate regime:** E_F is at least $3kT$ away from both band edges. In this case, carrier concentrations are low enough that Maxwell–Boltzmann statistics provide an accurate approximation.
- **Degenerate regime:** E_F moves closer than $3kT$ to a band edge (or into a band), i.e., $E_F > E_c - 3kT$ or $E_F < E_v + 3kT$. In this case, carrier concentrations are high and full Fermi–Dirac statistics must be used.

- (b) **Carrier density expressions (non-degenerate case)**

For $(E_v + 3kT) \leq E_F \leq (E_c - 3kT)$, the Boltzmann approximations are:

Electron density (n_0)

$$\begin{aligned}n_0 &= N_c e^{-\frac{E_c - E_F}{kT}}, \\n_0 &= n_i e^{\frac{E_F - E_i}{kT}}.\end{aligned}$$

Hole density (p_0)

$$\begin{aligned}p_0 &= N_v e^{-\frac{E_F - E_v}{kT}}, \\p_0 &= n_i e^{\frac{E_i - E_F}{kT}}.\end{aligned}$$

Problem 4 & 5: Intrinsic Concentrations

4. Calculate $n_i(1/cm^3)$ for intrinsic Si at $T = 300K$.
5. Calculate $n_i(1/cm^3)$ for intrinsic GaAs at $T = 300K$.

Solution

4. Intrinsic Si at $T = 300 K$

Parameters for Si:

$$\frac{m_n^*}{m_0} = 1.08, \quad \frac{m_p^*}{m_0} = 0.549, \quad E_G = 1.12 \text{ eV}.$$

Step 1: Calculate N_c and N_v

$$\begin{aligned} N_c &= 2.51 \times 10^{19} (1.08)^{3/2} \approx 2.82 \times 10^{19} \text{ cm}^{-3}, \\ N_v &= 2.51 \times 10^{19} (0.549)^{3/2} \approx 1.02 \times 10^{19} \text{ cm}^{-3}. \end{aligned}$$

Step 2: Calculate n_i

$$\begin{aligned} n_i &= \sqrt{(2.82 \times 10^{19})(1.02 \times 10^{19})} e^{-\frac{1.12}{2 \cdot 0.026}} \\ &= (1.70 \times 10^{19}) e^{-21.54} \\ &\Rightarrow \boxed{n_i \approx 7.73 \times 10^9 \text{ cm}^{-3}}. \end{aligned}$$

5. Intrinsic GaAs at $T = 300 K$

Parameters for GaAs:

$$\frac{m_n^*}{m_0} = 0.067, \quad \frac{m_p^*}{m_0} = 0.473, \quad E_G = 1.424 \text{ eV}.$$

Step 1: Calculate N_c and N_v

$$\begin{aligned} N_c &= 2.51 \times 10^{19} (0.067)^{3/2} \approx 4.35 \times 10^{17} \text{ cm}^{-3}, \\ N_v &= 2.51 \times 10^{19} (0.473)^{3/2} \approx 8.16 \times 10^{18} \text{ cm}^{-3}. \end{aligned}$$

Step 2: Calculate n_i

$$\begin{aligned} n_i &= \sqrt{(4.35 \times 10^{17})(8.16 \times 10^{18})} e^{-\frac{1.424}{2 \cdot 0.026}} \\ &= (1.88 \times 10^{18}) e^{-27.38} \\ &\Rightarrow \boxed{n_i \approx 2.41 \times 10^6 \text{ cm}^{-3}}. \end{aligned}$$

Problem 6

- (a) Determine the electron and hole density in Si when $(E_c - E_F) = 3kT$ at $T = 300\text{K}$.
 (b) Determine the electron and hole density in Si when $(E_F - E_v) = 3kT$ at $T = 300\text{K}$.

Solution

Given (Si at $T = 300\text{K}$):

$$n_i = 1.0 \times 10^{10} \text{ cm}^{-3}, \quad kT = 0.0259 \text{ eV}, \quad E_c - E_i \approx E_i - E_v \approx 0.56 \text{ eV}.$$

$$3kT = 3(0.0259 \text{ eV}) = 0.0777 \text{ eV}.$$

(a) **Case 1:** $E_c - E_F = 3kT$

$$\begin{aligned} E_F - E_i &= (E_c - E_i) - (E_c - E_F) = 0.56 - 0.0777 = 0.4823 \text{ eV}, \\ \frac{E_F - E_i}{kT} &= \frac{0.4823}{0.0259} \approx 18.63. \end{aligned}$$

$$\begin{aligned} n &= n_i e^{\frac{E_F - E_i}{kT}} = 10^{10} e^{18.63} \approx 1.1 \times 10^{18} \text{ cm}^{-3}, \\ p &= \frac{n_i^2}{n} \approx \frac{(10^{10})^2}{1.1 \times 10^{18}} \approx 9.0 \times 10^1 \text{ cm}^{-3}. \end{aligned}$$

$$\boxed{n \approx 1.1 \times 10^{18} \text{ cm}^{-3}, \quad p \approx 9.0 \times 10^1 \text{ cm}^{-3}}$$

(b) **Case 2:** $E_F - E_v = 3kT$

$$\begin{aligned} E_i - E_F &= (E_i - E_v) - (E_F - E_v) = 0.56 - 0.0777 = 0.4823 \text{ eV}, \\ \frac{E_i - E_F}{kT} &= \frac{0.4823}{0.0259} \approx 18.63. \end{aligned}$$

$$\begin{aligned} p &= n_i e^{\frac{E_i - E_F}{kT}} = 10^{10} e^{18.63} \approx 1.1 \times 10^{18} \text{ cm}^{-3}, \\ n &= \frac{n_i^2}{p} \approx \frac{(10^{10})^2}{1.1 \times 10^{18}} \approx 9.0 \times 10^1 \text{ cm}^{-3}. \end{aligned}$$

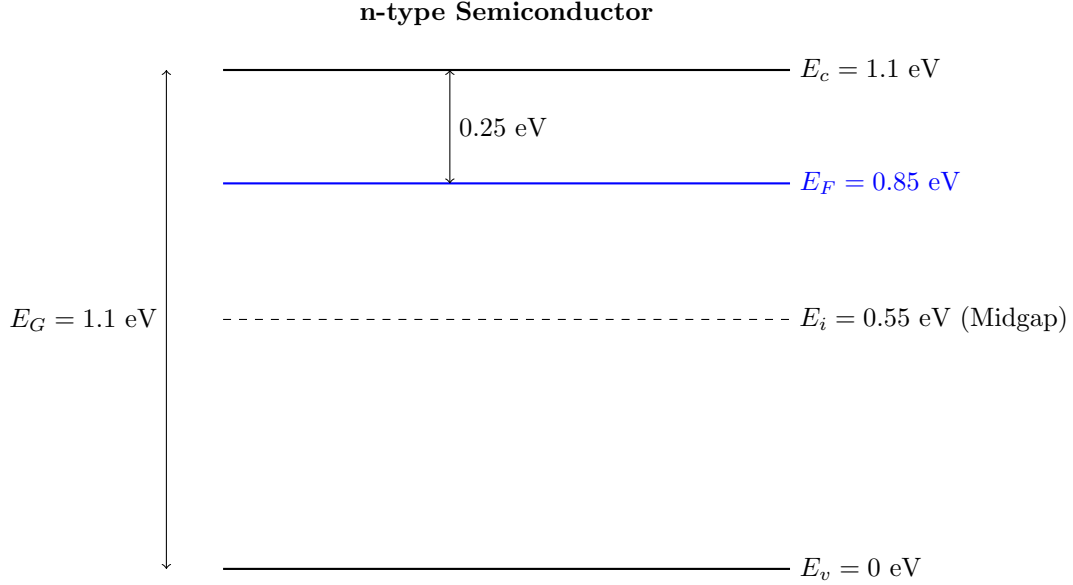
$$\boxed{p \approx 1.1 \times 10^{18} \text{ cm}^{-3}, \quad n \approx 9.0 \times 10^1 \text{ cm}^{-3}}$$

Problem 7

At $T = 300\text{K}$, a semiconductor has $E_G = 1.1\text{ eV}$ and $N_c = N_v$. Doped with $10^{15}/\text{cm}^3$ donors. E_F is 0.25 eV below E_c .

- Draw the band diagram ($E_v = 0$). Label E_c , E_F , and E_i .
- Calculate n , (c) Calculate n_i , (d) Calculate p .

Solution



- Band diagram:** shown above (with $E_v = 0$).
- Calculate n**

Assuming donors are fully ionized at room temperature,

$$n \approx N_d = 10^{15}\text{ cm}^{-3}.$$

- Calculate n_i**

Using

$$n = n_i e^{\frac{E_F - E_i}{kT}},$$

we have

$$E_F - E_i = 0.85\text{ eV} - 0.55\text{ eV} = 0.30\text{ eV}$$

and therefore

$$\begin{aligned}
 n_i &= \frac{n}{e^{\frac{E_F - E_i}{kT}}} = \frac{10^{15}}{e^{\frac{0.30}{0.026}}} = \frac{10^{15}}{e^{11.538}} \approx \frac{10^{15}}{1.025 \times 10^5} \\
 &\approx 9.75 \times 10^9\text{ cm}^{-3}.
 \end{aligned}$$

- Calculate p**

Using the law of mass action $np = n_i^2$,

$$p = \frac{n_i^2}{n} = \frac{(9.75 \times 10^9)^2}{10^{15}} = \frac{9.51 \times 10^{19}}{10^{15}} \approx 9.51 \times 10^4\text{ cm}^{-3}.$$

Problem 8

A semiconductor requires n-type material ($n \gg p$) at $T = 400\text{K}$. Use the provided n_i plots.

(a) Would As doped Si at $10^{15}/\text{cm}^3$ meet this criteria? Explain with equations.

(b) Would Sb doped Ge at $10^{15}/\text{cm}^3$ meet this criteria? Explain with equations.

Solution

(a) As-doped Si at $T = 400\text{ K}$

Step 1: Get n_i for Si at 400 K

From the plot on slide 33 (or standard reference):

$$n_i(\text{Si}, 400\text{ K}) \approx 3 \times 10^{12} \text{ cm}^{-3}$$

Step 2: Compare doping to intrinsic level

$$N_D = 10^{15} \gg n_i = 3 \times 10^{12},$$

so donors dominate and the material is in the extrinsic (n-type) regime.

Step 3: Compute carrier concentrations

$$\begin{aligned} n &\approx N_D = 10^{15} \text{ cm}^{-3}, \\ p &= \frac{n_i^2}{n} = \frac{(3 \times 10^{12})^2}{10^{15}} = 9 \times 10^9 \text{ cm}^{-3}. \end{aligned}$$

Step 4: Check criterion ($n \gg p$)

$$\frac{n}{p} \approx \frac{10^{15}}{9 \times 10^9} \approx 10^5.$$

Conclusion: YES — As-doped Si at 10^{15} cm^{-3} is strongly n-type at 400 K.

(b) Sb-doped Ge at $T = 400\text{ K}$

Step 1: Get n_i for Ge at 400 K

$$n_i(\text{Ge}, 400\text{ K}) \approx 1 \times 10^{15} \text{ cm}^{-3}.$$

Step 2: Compare doping to intrinsic level

$$N_D = 10^{15} \approx n_i,$$

so the material is **not** strongly extrinsic. We must use the exact charge neutrality calculation.

Step 3: Solve carrier concentrations

$$\begin{aligned} n &= \frac{N_D + \sqrt{N_D^2 + 4n_i^2}}{2} = \frac{10^{15} + \sqrt{10^{30} + 4(10^{30})}}{2} \approx 1.62 \times 10^{15} \text{ cm}^{-3} \\ p &= \frac{n_i^2}{n} = \frac{10^{30}}{1.62 \times 10^{15}} \approx 6.17 \times 10^{14} \text{ cm}^{-3}. \end{aligned}$$

Step 4: Check criterion ($n \gg p$)

$$\frac{n}{p} \approx \frac{1.62 \times 10^{15}}{6.17 \times 10^{14}} \approx 2.6.$$

Conclusion: NO — Sb-doped Ge at 10^{15} cm^{-3} is not strongly n-type at 400 K (n is only ~ 2.6 times larger than p).

Problem 9

Si is doped with 10^{15} atoms/cm³ of phosphorous. What are the electron and hole concentrations at $T = 540\text{K}$?

Solution

1. Determine n_i for Si at 540 K

Since 540 K is high, n_i will be significantly larger than at room temperature. We can scale n_i from 300 K.

$$n_i(T) \approx n_i(300) \left(\frac{T}{300} \right)^{1.5} \exp \left[-\frac{E_g}{2k} \left(\frac{1}{T} - \frac{1}{300} \right) \right]$$

Using $n_i(300) = 10^{10} \text{ cm}^{-3}$, $E_g = 1.12 \text{ eV}$, and $k = 8.617 \times 10^{-5} \text{ eV/K}$:

- T -factor: $(540/300)^{1.5} \approx (1.8)^{1.5} \approx 2.41$
- Exponent factor: $\frac{1.12}{1.72 \times 10^{-4}} \left(\frac{1}{300} - \frac{1}{540} \right) \approx 6511(0.00333 - 0.00185) \approx 6511(0.00148) \approx 9.63$
- $\exp(9.63) \approx 1.52 \times 10^4$

$$n_i(540) \approx 10^{10} \times 2.41 \times 1.52 \times 10^4 \approx 3.7 \times 10^{14} \text{ cm}^{-3}$$

2. Compare Doping vs Intrinsic Concentration

$$N_D = 10^{15} \text{ cm}^{-3}, \quad n_i \approx 3.7 \times 10^{14} \text{ cm}^{-3}.$$

Since n_i is comparable to N_D (only $\sim 3\times$ smaller), we cannot assume $n \approx N_D$. We must use the full quadratic solution derived from charge neutrality ($n = p + N_D$ and $pn = n_i^2$).

3. Calculate Electron Concentration (n)

$$n = \frac{N_D + \sqrt{N_D^2 + 4n_i^2}}{2}$$

Substitute values (units of 10^{14} for simplicity):

$$\begin{aligned} n &= \frac{10 + \sqrt{100 + 4(3.7)^2}}{2} \times 10^{14} \\ &= \frac{10 + \sqrt{100 + 54.76}}{2} \times 10^{14} \\ &= \frac{10 + 12.44}{2} \times 10^{14} \\ &= 11.22 \times 10^{14} = \mathbf{1.12 \times 10^{15} \text{ cm}^{-3}}. \end{aligned}$$

4. Calculate Hole Concentration (p)

$$p = \frac{n_i^2}{n} = \frac{(3.7 \times 10^{14})^2}{1.12 \times 10^{15}} = \frac{13.69 \times 10^{28}}{11.2 \times 10^{14}} \approx \mathbf{1.22 \times 10^{14} \text{ cm}^{-3}}.$$

Final Answer:

$n \approx 1.12 \times 10^{15} \text{ cm}^{-3}, \quad p \approx 1.22 \times 10^{14} \text{ cm}^{-3}$

Problem 10

Determine the intrinsic Fermi level of GaAs at:

(a) $T = 300\text{K}$

(b) $T = 400\text{K}$

Solution

The intrinsic Fermi level E_i is displaced from the midgap energy $E_{\text{mid}} = \frac{E_c + E_v}{2}$. The formula is:

$$E_i - E_{\text{mid}} = \frac{3}{4}kT \ln \left(\frac{m_p^*}{m_n^*} \right)$$

Parameters for GaAs (from Problem 5):

$$m_n^* = 0.067m_0, \quad m_p^* = 0.473m_0.$$

Calculate the logarithmic term:

$$\ln \left(\frac{m_p^*}{m_n^*} \right) = \ln \left(\frac{0.473}{0.067} \right) = \ln(7.06) \approx 1.954.$$

Thus, the shift is:

$$\Delta E = \frac{3}{4}(1.954)kT \approx 1.466 kT.$$

(a) **At $T = 300\text{ K}$:**

$$kT \approx 0.0259\text{ eV}.$$

$$E_i - E_{\text{mid}} = 1.466(0.0259) \approx \mathbf{0.038\text{ eV}}.$$

The intrinsic Fermi level is **0.038 eV above** the midgap.

(b) **At $T = 400\text{ K}$:**

$$kT = 0.0259 \times \frac{400}{300} \approx 0.0345\text{ eV}.$$

$$E_i - E_{\text{mid}} = 1.466(0.0345) \approx \mathbf{0.051\text{ eV}}.$$

The intrinsic Fermi level is **0.051 eV above** the midgap.